

TEMIR, Kazakhstan

WMO: 354260

Lat: 49.1422N

Lon: 57.1217E

Elev: 234

StdP: 98.55

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-28.2	-25.2	-31.6	0.2	-27.9	-28.7	0.3	-25.1	9.2	-11.5	8.2	-10.7	1.5	320	0.538

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.7	36.0	18.6	34.3	18.4	32.6	17.9	20.2	31.3	19.5	30.7	18.8	29.8	3.4	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.8	12.3	23.0	15.8	11.6	22.7	14.8	10.8	23.0	58.8	31.8	56.4	30.4	54.3	29.9	24.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.2	7.0	6.1	-31.1	38.7	3.9	1.8	-34.0	40.0	-36.3	41.1	-38.5	42.1	-41.4	43.4		
(5)			-31.1	21.5	4.4	1.2	-34.3	22.3	-36.9	23.0	-39.4	23.7	-42.6	24.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.4	-12.6	-11.6	-3.5	8.7	16.1	21.4	24.0	22.5	15.2	6.7	-2.0	-8.7		
	DBStd	14.15	6.61	6.88	6.20	5.79	4.55	4.38	3.67	4.41	4.60	4.89	5.89	6.30		
	HDD10.0	2901	701	605	421	91	6	1	0	0	9	126	361	580		
	HDD18.3	4826	959	838	678	290	99	19	3	15	114	360	611	838		
	CDD10.0	1603	0	0	1	54	194	342	434	388	165	24	1	0		
	CDD18.3	486	0	0	0	2	29	111	179	145	19	0	0	0		
	CDH23.3	6211	0	0	0	43	404	1442	2204	1807	303	7	0	0		
	CDH26.7	2922	0	0	0	4	132	679	1092	904	110	1	0	0		
(14) Wind	WSAvg	2.6	2.7	2.8	3.0	2.9	2.8	2.6	2.5	2.4	2.3	2.4	2.5	2.6		
(15) Precipitation	PrecAvg	273	24	21	19	27	25	23	25	19	17	24	29	26		
	PrecMax	352	71	71	56	83	78	76	82	86	80	71	58	70		
	PrecMin	158	2	0	1	0	0	0	0	0	0	0	2	1		
	PrecStd	54	16	15	13	21	20	18	20	16	15	15	14	17		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.8	1.6	15.6	27.1	32.2	37.2	38.2	38.1	34.5	23.8	12.3	3.8		
		MCWB	0.1	0.4	8.1	15.0	16.9	19.1	19.0	18.6	16.9	12.3	8.8	1.5		
	2%	DB	-0.2	0.4	10.5	24.4	30.0	35.1	35.9	35.7	29.5	20.8	9.5	1.4		
		MCWB	-0.8	-0.3	6.4	14.1	16.6	18.6	18.6	18.4	14.8	11.5	6.8	0.3		
	5%	DB	-1.5	-0.7	6.3	21.5	27.8	33.0	34.0	33.8	26.7	18.1	7.1	0.5		
		MCWB	-2.1	-1.4	3.6	12.4	16.1	18.1	18.4	18.2	14.0	9.9	5.0	-0.4		
	10%	DB	-3.0	-2.1	3.6	18.7	25.2	30.8	32.3	31.7	24.1	15.2	5.0	-0.8		
		MCWB	-3.7	-2.8	1.9	10.8	14.9	17.3	18.1	17.4	13.3	8.7	3.4	-1.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.2	0.7	9.3	16.1	19.6	22.0	21.2	20.7	17.8	14.1	10.4	2.0		
		MCDB	0.7	1.1	14.0	24.9	29.3	33.4	31.6	32.5	31.7	21.2	12.0	3.1		
	2%	WB	-0.8	-0.1	6.3	14.3	18.1	20.0	20.2	19.7	15.8	12.2	7.0	0.7		
		MCDB	-0.2	0.4	10.2	22.9	26.8	30.7	31.2	32.0	26.6	18.9	8.8	1.3		
	5%	WB	-2.1	-1.3	3.8	12.9	16.7	19.0	19.6	18.9	14.8	10.7	5.1	-0.4		
		MCDB	-1.6	-0.7	6.2	20.4	25.1	30.0	30.5	31.2	24.4	16.5	6.9	0.4		
	10%	WB	-3.7	-2.8	2.0	11.4	15.4	18.1	19.0	18.1	13.8	9.2	3.5	-1.6		
		MCDB	-3.1	-2.1	3.4	17.9	23.5	28.5	29.7	29.5	22.9	14.6	4.9	-0.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.5	8.2	8.1	11.2	12.5	13.3	12.7	13.4	13.4	11.3	6.9	6.7		
		MCDBR	5.8	6.0	10.4	15.2	16.1	15.9	15.0	15.6	17.3	16.1	9.3	6.2		
		MCWBR	5.4	5.6	6.9	7.4	6.2	5.5	4.7	5.5	7.2	8.8	6.6	5.4		
		MCDBR	5.6	5.5	9.7	14.1	14.1	14.1	13.1	14.2	14.9	14.0	8.1	5.6		
	5% WB	MCWBR	5.6	5.3	6.7	7.3	6.2	5.6	4.7	5.3	6.8	8.4	6.3	5.1		
(40) Clear-Sky Solar Irradiance	taub		0.303	0.329	0.392	0.436	0.422	0.417	0.412	0.422	0.370	0.357	0.337	0.308		
	taud		2.084	2.020	1.985	2.048	2.172	2.248	2.275	2.220	2.319	2.277	2.210	2.114		
	Ebn at Noon		712	786	796	807	839	845	843	813	824	760	674	647		
	Edh at Noon		87	118	147	154	142	132	127	128	105	92	78	74		
(44) All-Sky Solar Radiation	RadAvg		1.45	2.65	3.73	4.79	6.12	6.78	6.56	5.71	4.25	2.43	1.16	0.93		
	RadStd		0.20	0.21	0.46	0.49	0.54	0.48	0.36	0.35	0.28	0.31	0.14	0.18		

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(47) Regional (4 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		N/A	N/A	N/A	+1.10	N/A	N/A	N/A	+105	+84

Nomenclature: See separate page